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EDCEW-LLM: Error detection and correction in English writing: A large language model-based approach

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ABSTRACT

This study investigates the use of large language models (LLMs) for error detection and correction in English writing (EDCEW), focusing on the challenges of overcorrection and pedagogical feedback. We explore the relationship between targeted prompting strategies, transformer model architectures, and the error profiles of English language learners across CEFR proficiency levels (A, B, and C). We evaluated LLM through zero-shot, few-shot and fine-tuning approaches, showing that overcorrection is more common in texts from advanced learners, while lower proficiency levels show unique patterns of errors. To address this issue, we introduce a two-stage pipeline: first, a fine-tuned LLM leverages sequence-to-sequence modeling to perform precise token-level error detection and correction, achieving competitive F1 scores across multilingual datasets (English, German, Chinese); second, a prompted LLM (GPT 3.5) generates rule-based, natural language explanations for each identified error, enhancing the educational values. We present comprehensive benchmark results on standard WEDC datasets (BEA-2019, JFLEG), supported by human evaluations, that show the effectiveness of the proposed approach in producing accurate corrections and linguistically informative error explanations. The proposed model improves correction accuracy and facilitates deeper linguistic comprehension through explicit, pedagogically sound feedback, addressing a significant limitation of existing automated GEC tools. Moreover, this research advances LLM-driven WEDC by providing a robust framework that balances correction precision with pedagogical utility. Qualitative results further highlight the model's ability to address diverse error types, such as Subject+Verb agreement, redundant comparisons, and incorrect preposition usage, with contextually appropriate corrections.

1. Introduction

The increasing reliance on written communication in both personal and professional contexts has underscored the importance of clear and accurate writing. As language proficiency becomes increasingly critical, the need for effective English writing error detection and correction tools has become evident [1]. With the rise of digital communication platforms, automated content generation systems, and educational technologies [2], the demand for advanced systems capable of identifying and correcting errors in written English has surged. The ability to detect and correct writing errors is crucial for improving communication, enhancing the linguistic capabilities of non-native speakers and fostering language proficiency.

Despite significant advancements in Natural Language Processing (NLP) and Artificial Intelligence (AI), automatic English writing error detection and correction remains a challenging task, primarily due to the complexity of language itself [3]. Like many other natural languages such as Chinese, Spanish, German, etc., English is naturally ambiguous and context-sensitive. Due to this reason, finding syntactic,

semantic, and lexical mistakes is a challenging task [4]. There is no doubt that deep learning (DL), especially transformer-based models including BERT (Bidirectional Encoder Representations from Transformers) and GPT-3, advanced error detection and correction to a certain extent. However, there are still issues with accurate error detection, particularly when it comes to contextual understanding, word choice, and grammar correction [5,6].

Scalability and adaptability issues hinder traditional rule-based error detection systems. These systems frequently depend on predetermined rules, such word lists or grammar rules, which are unable to keep up with the complexity and diversity of writing styles, a wide range of vocabulary, and changing linguistic trends [7]. However, even though statistical techniques and machine learning (ML) models are more flexible, they still struggle to distinguish between grammatically correct variations and actual errors. This case is more common when dealing with complex sentence structures, colloquial expressions, and cross-linguistic influences.

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These problems have been addressed in large part by recent developments in DL, particularly in artificial neural network (ANN) architecture [8,9]. Because they provide more robust and context-sensitive models, neural-based techniques — specifically, sequence-to-sequence models, attention mechanisms, and transformers — are being used more and more for mistake detection and correction [10]. This represents a substantial departure from the conventional, rule-based approaches that frequently overlooked the subtleties of normal language usage [11]. These methods not just identify errors but also suggest contextually effective corrections.

More adequate datasets for the identification and correction of errors have gained prominence within the realm of science. Publicly available datasets, such as the first certificate in English (FCE) [12], Lang-8 [13], and NUS [14], offer extensive annotated data, supporting the establishment of advanced algorithms. Advanced ML-based models are trained using these datasets and are used for grammatical error identification and correction. Nevertheless, despite these advances, certain challenges persist, one of which is the models' ability to generalize across various writing domains and proficiency levels. In practical writing contexts, where styles vary substantially, or when collaborating with non-native English writers who might display errors influenced by their original languages, models trained on specific data sets might exhibit suboptimal performance [15]. In addition, interpretability, the capability for models to support the choices, remains an essential topic of inquiry. Building trust in these systems, particularly in sensitive contexts such as academic writing or professional communication, requires understanding why a model identified a specific portion of text as erroneous and how it determined the appropriate correction.

Moreover, there is a growing need to focus on error context to better differentiate between acceptable variations and errors. For instance, the distinction between regional linguistic variations and genuine grammar mistakes needs to be carefully considered in error detection systems. The research community has yet to achieve a universally acceptable method for error contextualization, which limits the effectiveness of existing systems. This research aims to develop a novel approach to English writing error detection and correction, utilizing state-of-the-art DL techniques, specifically transformer-based models such as BERT and GPT-3, while integrating the strengths of contextual learning and error classification [16,17]. By utilizing a context-aware approach, we aim to improve the accuracy and adaptability of error detection systems across diverse writing styles and proficiency levels. Additionally, we propose innovations in explainability and user feedback integration, allowing for more transparent and interpretable correction suggestions.

The contribution of this paper is twofold. First, we aim to develop a model that not only detects grammatical, lexical, and syntactical errors but also understands the context in which these errors occur to provide contextualized corrections. Second, by focusing on the explainability of model predictions, we seek to increase the transparency of the error correction process, ensuring that the end-user can trust and understand the suggestions made by the system. The contributions of this paper are summarized as follows:

We propose a novel two-stage learning framework for English Writing Error Detection and Correction (WEDC) that LLMs for precise error detection and contextually relevant corrections.

We introduce a proxy perplexity-based error detection mechanism that utilizes an ensemble of LLMs to identify statistical irregularities in text, enabling robust error detection across diverse writing styles and proficiency levels. Moreover, we develop a generative LLM-based error correction framework that not only corrects errors but also provides natural language explanations, enhancing the pedagogical value of the system.

We demonstrate the effectiveness of our approach through comprehensive evaluations on standard WEDC datasets (BEA-2019, JFLEG), achieving state-of-the-art performance in terms of precision, recall, F1-score, and human evaluations. Furthermore, we

evaluate the robustness of our model under various perturbations and demonstrate the impact of knowledge distillation in improving computational efficiency without significant loss in performance.

The remaining paper is structured in the following order. In Section 2, we review the current state of English writing error detection and correction and delve into existing methods and their limitations. Section 3 illustrates the proposed framework of EDCER using LLMs. In Section 4, we present the error detection using ML classification techniques. Section 5 talks about the experimental results of the model, focusing on quantities as well as qualitative results. In the last section (Section 6), we conclude the paper with a summary of our findings and future directions for research in this area.

2. Related work

In this decade, NLP has concentrated on identifying and rectifying faults in English text. The past few years have witnessed significant advancements in this domain, particularly with the emergence of neural network models. The current advanced systems are mostly based on statistical methods such as ML and DL to detect and correct writing errors, whereas the conventional approaches were mostly using rule-based and lexicon-driven techniques [18–20]. These issues encompass lexical inaccuracies, spelling errors, typos in grammar, and more complex semantic inconsistencies. This section provides an in-depth review of the latest developments, key approaches, and limitations in the domain of English writing error detection and correction.

2.1. Rule-based approaches: Foundations of error detection and correction

Initial error correction methods predominantly employed rule-based approaches. These methods were designed using predefined language frameworks or manually created grammatical regulations. The initial models included grammar and correction tools that concentrated on grammar, punctuation, spelling, and fundamental grammatical structures [21]. For instance, the spelling checking and correction module in Microsoft Office [22,23]. These systems employed dictionaries and basic regular expressions to recognize errors. While these systems excelled at recognizing superficial issues, they often struggled with more complex language phenomena such as disambiguation of word meaning and ambient grammatical errors [24,25].

With advancements in computational power and data availability, the field increasingly shifted towards statistical methodologies. By analyzing the likelihood of certain word combinations using large sets of language data, these methods helped computers better recognize complicated grammar problems and find mistakes based on context instead of strict rules. N-gram models are widely used for checking grammar, helping to spot wrong word choices and arrangements by predicting how likely a word or phrase is to appear in a certain context [26].

2.2. Machine learning-based approaches

The advent of ML methods in identifying errors marked a significant transition from rule-driven to data-driven approaches. The predominant early ML models were designed to ascertain whether a specific text contained an error, focusing mostly on classification problems. [27]. To discern between correct and incorrect sentence structure, methods including LR, DT, SVM, etc., were applied. These algorithms utilized many features, including word frequency ranges, syntactic patterns, and POS tags, to detect grammatical errors.

The development of mistake-specific correction methods, where investigators train separate models to address certain error types, such as Subject+Verb concordance, conjunction misuse, or word errors, represented a notable advancement in the ML era [28]. These methods enable computers to be more focused and excel at certain jobs; however, they sometimes lack generality across many types of writing errors. Although ML models surpassed traditional rule-based methods, they still failed to completely incorporate the intricacies of language.

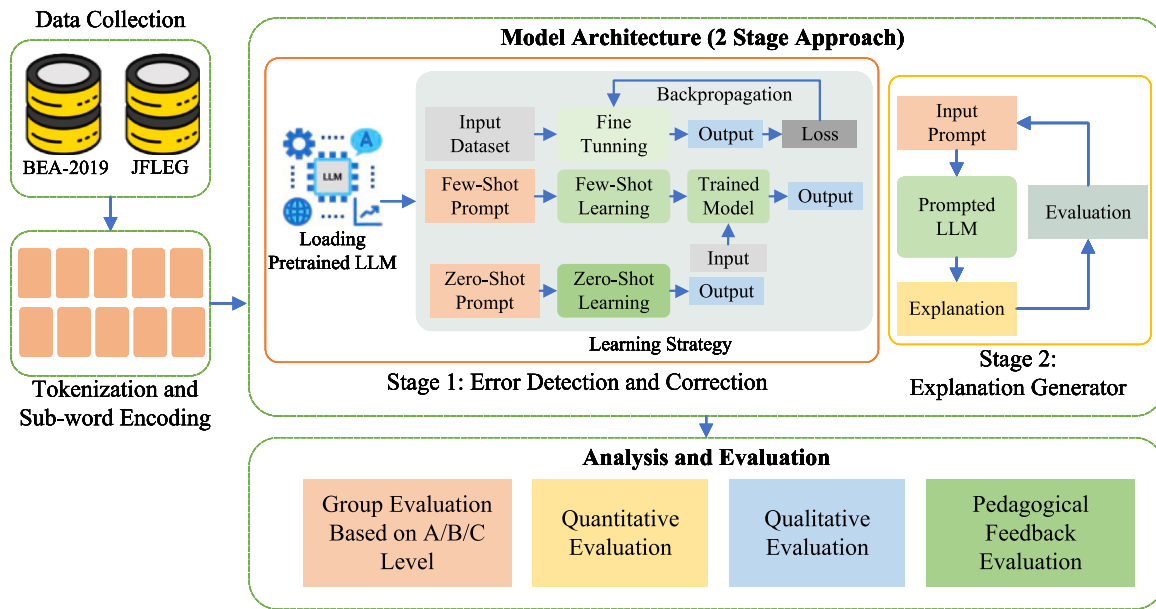


Fig. 1. Proposed methodology for English writing error detection and correction using large language models.

2.3. Deep learning-based approaches

The advancement of DL models has drastically altered error identification and correction research during the past decade. Neural networks' capacity to articulate sequential links in language has given rise to their prominence, particularly RNN-based models and more advanced variants, such as LSTMs [3,29]. These models exhibit an aptitude to acquire contextual understanding, which is crucial for detecting errors in increasingly complex sentence structures. LSTMs are especially adept at error detection tasks that necessitate an understanding of syntactic structure and context, as they are optimally designed for NLP problems with dependency chains.

Concurrently, the domain has experienced a transformation because of encoder–decoder architectures [30], especially those based on the Transformer paradigm. BERT, along with its derivatives like GPTs, has demonstrated outstanding effectiveness in language understanding. The model can assess the entire sentence to detect context-dependent errors due to BERT's bidirectional contextual comprehension. Moreover, GPT-3 and related transformer paradigms excel at contextual error correction, an area where traditional systems are inadequate, as they generate modifications based on the context of the input [31,32].

By employing refined pre-trained models like BERT for tasks like grammar checking and sentence repair, recent research has effectively used transformer-based models for English writing mistake correction. These models beat previous statistical and machine-learning techniques in terms of accuracy and resilience because they learn the subtleties of real language as well as the underlying grammatical rules from large datasets. The Transformer architecture is very successful for error detection that necessitates comprehension of the sentence context because it uses attention methods to enable these models to concentrate on the most pertinent portions of a phrase [33,34].

2.4. Hybrid models and transfer learning

A noteworthy advancement in contemporary studies has been the application of hybrid models, which integrate conventional techniques with DL [35]. In order to translate incorrect sentences into correct ones, researchers have experimented with neural machine translation (NMT)-based techniques that frame sentence-level error correction as

a translation task [36,37]. By combining the flexibility of statistical methods with the resilience of DL models, these hybrid models enhance error detection and correction across a range of writing styles [38].

Transfer learning, in which models pre-trained on extensive, generic corpora are refined on particular error detection tasks, is another area of notable progress [39]. This technique has been particularly beneficial in cases where annotated data for specific writing errors is scarce. For example, pre-trained models such as BERT, GPT-2, and RoBERTa have been adapted to identify and correct errors in specific contexts, such as academic writing or non-native English writing, by fine-tuning the models on relevant datasets [40,41].

2.5. Contextualized error detection and correction

One of the main limitations of earlier models was their inability to effectively handle contextual errors—where the correct usage of words, phrases, or structures depends heavily on the surrounding context [42, 43]. Contextualized error detection requires systems to differentiate between valid variations of grammar (e.g., the use of both the singular and plural forms) and errors that compromise clarity or grammatical correctness. Recent developments in contextual embeddings have tackled this challenge by representing words and phrases in context-dependent embeddings [44]. Transformer-based architectures such as BERT and XLNet leverage these embeddings to ensure that the model can better understand the relationships between words in a sentence, leading to more accurate error detection and correction [45,46]. Moreover, semantic errors, such as incorrect word choices or the use of non-native expressions, are difficult to detect with traditional methods [45,47]. Current transformer-based models are significantly better at understanding semantic consistency, and studies have demonstrated their effectiveness in correcting not just grammatical mistakes but also errors related to word sense disambiguation, word collocation, and preposition usage. While the field has made tremendous progress, several challenges remain. One key limitation is the generalization ability of DL models. Models trained on specific datasets often perform well within the training domain but struggle when applied to diverse writing styles or non-native English speakers [48,49]. Furthermore, current models are still unable to provide an explanation for their choices, which restricts their use in fields like education or business where

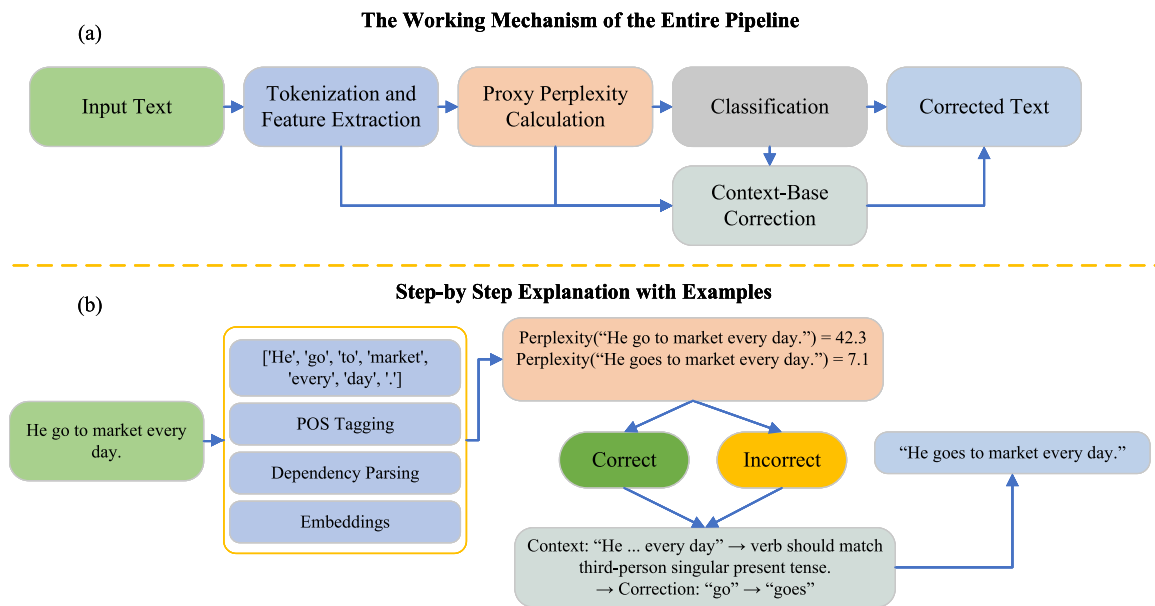


Fig. 2. Complete pipeline of the proposed methodology.

openness is essential. Another barrier is still the requirement for more varied and annotated datasets. Despite their widespread use, databases like Lang-8, FCE, and the NUS Corpus of Learner English only cover a small portion of possible writing styles, and they do not adequately reflect the range of mistakes that occur in real-world writing [46,50]. To further model development, more extensive datasets reflecting a wider variety of writing contexts, from corporate communications to creative writing, must be created.

3. Proposed methodology for English writing error detection and correction using large language models

It is the revolutionary potential of LLMs that motivates our suggested model for WEDC. Nuanced contextual comprehension, natural and fluid corrections, managing a variety of mistake kinds, and adjusting to different writing styles and learner competence levels are all areas where current WEDC systems frequently fall short. With training on large datasets, LLMs can produce writing that is human-like, model intricate language patterns, and perhaps overcome these drawbacks to identify and correct errors more accurately and contextually. Significant advantages are provided by these developments for professional communication, academic support, and language instruction.

Notwithstanding the enormous potential of LLMs in WEDC, a number of important issues need to be resolved. Among these is the tendency for overcorrection, which occurs when LLMs needlessly change material that is correct, especially when it comes to advanced learners. Unfair or discriminating error detection and rectification results may result from biases present in the training data. Many LLM-based systems lack explainability, which prevents users from understanding the rationale for corrections and impedes the learning process. Large model deployment and training may provide a practical problem due to the significant computational resources needed. Our suggested approach seeks to successfully negotiate the continuous challenges of defining what qualifies as a stylistic choice vs. a “error” and guaranteeing robustness against informal or loud language.

To address these challenges, we propose a two-stage framework that leverages the strengths of LLMs while mitigating their limitations as shown in Fig. 1. The first stage focuses on error detection, where we utilize an ensemble of LLMs to compute proxy perplexity scores for the input text. This approach allows us to identify potential errors based

on statistical irregularities in the text. The second stage involves error correction, where we employ a generative LLM to produce contextually relevant corrections and explanations for the detected errors. By combining these two stages, we aim to create a robust and efficient WEDC system that not only identifies and corrects errors but also provides valuable feedback to users.

To further highlight the proposed methodology, we present a detailed overview of the pipeline in Fig. 2. The pipeline consists of two main components: the overall workflow and the explanation. The upper part of the figure illustrates the complete pipeline, while the lower part provides a detailed explanation of each step involved in the process through examples.

3.1. Tokenization and feature extraction

The first stage of our methodology is tokenization of the input text. Given an input sentence, $S = [w_1 w_2 \dots w_t]$ where w_i is the i th word in the sequence, and what t is the length of the sentence? Tokenization transforms the sentence into a sequence of tokens that the model can process. We employ a subword tokenization algorithm, specifically Byte-Pair Encoding (BPE), which addresses challenges like out-of-vocabulary words by breaking them into frequently occurring subword units [51–53]. BPE iteratively merges the most frequent adjacent character pairs in the training data to build a vocabulary of these subwords.

For example, the word “unambiguously” might be tokenized into [“un”, “ambi”, “guous”, “ly”]. This allows the model to handle unseen words and capture morphological information. Following tokenization, if the sentence S is transformed into a token sequence $[tok_1 tok_2 \dots tok_m]$, each token tok_i is then converted into a dense, continuous vector representation known as an embedding. The embedding process can be mathematically represented as:

$$E(tok_i) = W_e \cdot O(tok_i) + b_e \quad (1)$$

where $\forall i \in \{1 2 \dots m\}$, $O(tok_i) \in \mathbb{R}^v$ is the one-hot vector representation of the token tok_i , $W_e \in \mathbb{R}^{d \times v}$ is the embedding weight matrix, $b_e \in \mathbb{R}^d$ is the bias vector, and $E(tok_i) \in \mathbb{R}^d$ is the resulting embedding vector of dimension d . These embeddings capture semantic and syntactic properties of the token learnt during the LLM’s pre-training. They serve as the fundamental input features for the subsequent layers of the LLM, enabling it to understand and process the linguistic information

Algorithm 1 Mathematical Framework for Text-to-Perplexity Conversion using n-gram Transformation

Require: Input text T , n-gram size n , Pre-trained language model $\mathcal{M}$

Ensure: Perplexity vector $= [P_1 \ P_2 \ \dots \ P_k]$

1: **Initialize:** $[\]$ {Empty perplexity vector}

2: **Step 1: Tokenization**

3: $tokens \leftarrow \text{TOKENIZE}(T)$ {Convert text T into a sequence of tokens $[w_1 \ w_2 \ \dots \ w_t]$ }

4: **Step 2: N gram Transformation**

5: $grams \leftarrow \{g_i = [w_i \ w_{i+1} \ \dots \ w_{i+n-1}] \mid i \in [1 \ t - n + 1]\}$ {Generate n-grams g_i from tokens}

6: **Step 3: Perplexity Calculation**

7: **for all** $g_i \in grams$ **do**

8: Compute the conditional probability $p_{\mathcal{M}}(g_i) = \prod_{j=1}^n p_{\mathcal{M}}(w_{i+j-1} \mid w_i \ \dots \ w_{i+j-2})$

9: Compute perplexity for g_i :

$$P_i = \exp \left(-\frac{1}{n} \sum_{j=1}^n \log p_{\mathcal{M}}(w_{i+j-1} \mid w_i \ \dots \ w_{i+j-2}) \right)$$

10: $\text{vector} \leftarrow \text{APPEND}(P_i)$ {Append perplexity P_i to vector }

11: **end for**

12: **return**

13: **Helper Functions:**

14: $\text{TOKENIZE}(T)$: Split text T into tokens $[w_1 \ w_2 \ \dots \ w_t]$.

15: $\text{GENERATEGRAMS}(tokens \ n)$: Generate n-grams g_i as:

$$g_i = [w_i \ w_{i+1} \ \dots \ w_{i+n-1}] \quad i \in [1 \ t - n + 1]$$

within the input sentence for error detection and correction [54–56].

3.2. Perplexity calculation

Perplexity, a measure of the pre-trained model's surprise with the input, is the most commonly used model-based evaluation of text in natural language processing and modeling. Because the perplexity of a text is calculated as the product of the probabilities of individual tokens, it is possible that one unlikely token significantly reduces the probability (i.e., increases the perplexity) of an otherwise highly probable input, despite being a simple typographical error. The computation of perplexity, which is a scalar value that corresponds to the full input, does not include information about the probability distribution. A pretty decent text with one unlikely token and another text with equally likely tokens might both have the same perplexity rating, especially for lengthier texts. Perplexity P is a standard measure used to evaluate the performance of language models. It quantifies how well a language model predicts a sequence of words. Given a text $S = [w_1 \ w_2 \ \dots \ w_t]$, the true perplexity for a model $\mathcal{M}$ is defined as:

$$P_{\mathcal{M}}(S) = \exp \left(-\frac{1}{t} \sum_{i=1}^t \log p_{\mathcal{M}}(w_i \mid w_1 \ w_2 \ \dots \ w_{i-1}) \right) \quad (2)$$

where $p_{\mathcal{M}}(w_i \mid w_1 \ w_2 \ \dots \ w_{i-1})$ is the conditional probability assigned by the model $\mathcal{M}$ to the i th word given the previous words. Perplexity provides a scalar measure of the predictive performance of the model. However, it does not provide information about the specific errors present in the text. To address this, we used the n-gram to get a vectorize representation of the perplexity across different segments of the input text, the complete process of which is described in Algorithm 1.

In this study, we are interested in using perplexity as a proxy for error detections. The idea is that a well-trained language model should assign lower perplexity to grammatically correct and fluent text compared to text with errors. Therefore, we can use the perplexity score to identify potential errors in the input text.

3.3. Proxy perplexity calculation

Instead of directly computing the perplexity, we utilize next-token probability sampling based on pre-constructed n-gram dictionaries. The proxy perplexity $\downarrow$ or a sequence S on a model $\mathcal{M}$ can be defined as:

$$\hat{P}_{\mathcal{M}}(S) = \exp \left(-\frac{1}{t} \sum_{i=1}^t \log p_{\mathcal{M}}(w_i \mid n - G \ w_1 \ \dots \ w_{i-1}) \right) \quad (3)$$

where $n - G \ w_1 \ \dots \ w_{i-1}$ refers to the context (previous words) modeled by an n-gram (2-g, 3-g, or 4-g) from the dictionary for the corresponding model.

Proxy perplexity approximates the model's ability to predict a sequence using precomputed n-gram probabilities, providing an approximation of the true perplexity without needing to access the model directly.

3.4. Detection mechanism

Our proposed error detection mechanism leverages the principle that erroneous or less natural text is likely to exhibit a higher degree of statistical irregularity when evaluated by a well-trained language model compared to grammatically correct and fluent text. To operationalize this, we compute the proxy perplexity, as defined in the previous subsection, for a given input text across an ensemble of language models. This ensemble includes our target model $\mathcal{M}$, which we aim to evaluate for its error detection capabilities, and one or more baseline models trained on corpora of high-quality, human-generated English text. The underlying rationale is that a language model trained on predominantly correct text will assign higher probabilities (and thus lower perplexity) to sequences that align with the statistical patterns it has learned from that data. Conversely, when presented with text containing errors (grammatical, syntactic, or stylistic), the probability assigned by such a model to the erroneous segments is expected to be lower, leading to a higher overall perplexity score for the entire text [57–59].

By calculating the proxy perplexity $\hat{P}_{\mathcal{M}_i}(S)$ for the input text S with respect to each model $\mathcal{M}_i$ in our ensemble, we can establish a comparative measure of the text's statistical likelihood under different language models. Our core hypothesis is that if the input text contains a significant number of errors, its proxy perplexity will be comparatively higher when evaluated by models trained on correct human-generated text than when evaluated by a model that may have been exposed to or is more tolerant of such errors. Therefore, our detection mechanism posits that the model yielding the *highest* proxy perplexity for a given input text is the one under which the text appears statistically most anomalous or least predictable. In the context of error detection, we interpret this to mean that the text is less likely to have originated from a source that adheres to the linguistic patterns captured by that high-perplexity model, suggesting the presence of deviations from standard English writing [53,60,61].

Formally, for an input text S and an ensemble of models $\{\mathcal{M}_1 \ \mathcal{M}_2 \ \dots \ \mathcal{M}_n\}$, where $\mathcal{M}_1$ is our target model and $\mathcal{M}_2 \ \dots \ \mathcal{M}_n$ are baseline models trained on human-generated text, the error detection decision can be expressed as:

$$\mathcal{M}_{\max} = \arg \max_{\mathcal{M}_i \in \{\mathcal{M}_1 \ \mathcal{M}_2 \ \dots \ \mathcal{M}_n\}} \hat{P}_{\mathcal{M}_i}(S) \quad (4)$$

where $\mathcal{M}_{\max}$ is the model that assigns the highest proxy perplexity to the input text S . A significantly higher proxy perplexity for S under a baseline model compared to the target model $\mathcal{M}_1$ indicates a higher likelihood of errors within the text.

This comparative analysis of proxy perplexity across different mod-

Algorithm 2 LightGBM-Based Classification for Error Detection

Require: Proxy perplexity features $F = [\hat{P}_{\mathcal{M}_1} \ \hat{P}_{\mathcal{M}_2} \ \dots \ \hat{P}_{\mathcal{M}_c}]$, Labels $y \in \{0, 1\}$, Learning rate η , Number of boosting rounds N , Maximum tree depth d , Regularization parameter λ

Ensure: Predicted label $\hat{y}$ for input text

- 1: Initialize LightGBM parameters: learning rate η , maximum depth d , number of leaves, and loss function $\ell(y, \hat{y}) = -\frac{1}{n} \sum_{i=1}^n [y_i \log \hat{y}_i + (1 - y_i) \log (1 - \hat{y}_i)]$.
- 2: Input training dataset (F, y) .
- 3: Initialize predictions $\hat{y}_i^{(0)} = 0$ for all $i \in \{1, 2, \dots, n\}$.
- 4: **for** $t = 1$ to N **do**
- 5: Compute gradients $g_i^{(t)}$ and Hessians $h_i^{(t)}$ for each sample i :

$$g_i^{(t)} = \frac{\partial \ell(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)}} = \hat{y}_i^{(t-1)} - y_i$$

$$h_i^{(t)} = \frac{\partial^2 \ell(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)2}} = \hat{y}_i^{(t-1)} (1 - \hat{y}_i^{(t-1)}).$$

- 6: Construct a decision tree $T^{(t)}$ with maximum depth d :
 - 1) For each split, select the feature F_j that minimizes the loss reduction:

$$\Delta = \frac{(\sum_{i \in \text{left}} g_i^{(t)})^2}{\sum_{i \in \text{left}} h_i^{(t)} + \lambda} + \frac{(\sum_{i \in \text{right}} g_i^{(t)})^2}{\sum_{i \in \text{right}} h_i^{(t)} + \lambda} - \frac{(\sum_{i \in \text{all}} g_i^{(t)})^2}{\sum_{i \in \text{all}} h_i^{(t)} + \lambda}.$$

- 2) Assign leaf values v_{leaf} using:

$$v_{\text{leaf}} = -\frac{\sum_{i \in \text{leaf}} g_i^{(t)}}{\sum_{i \in \text{leaf}} h_i^{(t)} + \lambda}.$$

- 7: Update predictions $\hat{y}_i^{(t)}$ for each sample i :

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta \cdot T^{(t)}(F_i).$$

- 8: **end for**

- 9: Output final predictions $\hat{y}_i = \sigma(\hat{y}_i^{(N)})$, where $\sigma(x) = \frac{1}{1+e^{-x}}$ is the sigmoid function.

els forms the foundation of our proposed error detection mechanism.

4. Error detection model: Machine learning classification

Once we calculate the proxy perplexity, we input it into a classification model to distinguish between human-written and machine-generated content. Let $F = [\hat{P}_{\mathcal{M}_1} \ \hat{P}_{\mathcal{M}_2} \ \dots \ \hat{P}_{\mathcal{M}_c}]$ be the feature vector representing the proxy perplexity for each model $\mathcal{M}_1 \ \mathcal{M}_2 \ \dots \ \mathcal{M}_c$ where c represents the number of models used for detection. The classification model (say, LightGBM) is then trained to distinguish between these features [62]. Let y represent the binary label: $y = 1$ if the text is human-generated, and $y = 0$ if machine-generated. The output of the classifier can be represented as:

$$\hat{y} = \sigma(W_c F + b_c) \quad (5)$$

where $\hat{y}$ is the predicted label (human or machine).

Algorithm 2 describes the LightGBM-based classification process for error detection. The basic purpose of this algorithm is to train a LightGBM model on the proxy perplexity features and then use it to classify the input text as either human-generated or machine-generated. We need this classification step to identify the text that requires correction. The algorithm iteratively builds decision trees to minimize the classification error, leveraging gradient and hessian computations for optimization.

4.1. Contextual understanding for correction

Once errors are detected, we proceed to correct them. The core challenge in error correction is contextual understanding, which requires generating grammatically correct and contextually relevant text. We employ LightGBM for its ability to handle high-dimensional, tabular features with strong generalization and interpretability. It is trained on labeled examples of both human and LLM-generated texts using proxy perplexity vectors as input.

Let the input text with detected errors be denoted as $S_{err} = [w_1 \ w_2 \ \dots \ w_l]$. A generative language model (like GPT-4 or T5) will generate a corrected sentence $\hat{S} = [\hat{w}_1 \ \hat{w}_2 \ \dots \ \hat{w}_l]$, where each corrected token $\hat{w}_i$ is generated based on the previous tokens $[w_1 \ w_2 \ \dots \ w_{i-1}]$. The probability of generating the corrected sentence $\hat{S}$ given the erroneous sentence S_{err} is:

$$p_{\mathcal{M}}(\hat{S} | S_{err}) = \prod_{i=1}^l p_{\mathcal{M}}(\hat{w}_i | w_1 \ w_2 \ \dots \ w_{i-1}) \quad (6)$$

where $p_{\mathcal{M}}(\hat{w}_i | w_1 \ w_2 \ \dots \ w_{i-1})$ represents the conditional probability of generating the i th word $\hat{w}_i$ given the context of the previously generated words.

4.2. Ranking and selection of corrections

In the case of multiple possible corrections for an error, a ranking function $R(\hat{S})$ can be applied to evaluate the fluency and accuracy of the correction. This ranking function is based on perplexity (lower perplexity suggests better fluency) and contextual compatibility (higher likelihood of fitting into the original meaning).

The final ranked correction can be determined by:

$$R(\hat{S}) = \arg \max_{\hat{S}} (\text{Perplexity}(S_{err}) - \alpha \cdot \text{SD}(S_{err}, \hat{S})) \quad (7)$$

where α is a hyperparameter controlling the trade-off between perplexity and semantic distance, and $\text{SD}(S_{err}, \hat{S})$ measures how close the corrected sentence is to the original meaning (e.g., cosine similarity between embeddings of S_{err} and $\hat{S}$).

4.3. Knowledge distillation for efficiency

To ensure that the model is efficient, we propose knowledge distillation, where a smaller model $\mathcal{M}_{small}$ learns to replicate the behavior of a larger, pre-trained model $\mathcal{M}_{large}$. The loss function for distillation is defined as:

$$L_{\text{dis}} = \lambda \cdot D_{KL}(p_{\mathcal{M}_l}(\hat{S}) \| p_{\mathcal{M}_s}(\hat{S})) + (1 - \lambda) \cdot \text{task}(S_{err}, \hat{S}) \quad (8)$$

where D_{KL} is the Kullback–Leibler divergence between the probability distributions of the large and small models, and task is the task-specific loss (e.g., cross-entropy for error correction).

4.4. Robustness to perturbations

To test the robustness of the model to text perturbations (such as word deletions or temperature changes in the generation process), we evaluate the model's performance on perturbed datasets and use the accuracy measure under perturbations:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \quad (9)$$

where a perturbed text $S_{\text{perturbed}}$ is generated by introducing random deletions or variations, and the model's ability to detect and correct errors under such conditions is evaluated.

Table 1

Dataset statistics.

Dataset	Total sentences	Avg. words/Sentence	Avg. errors/Sentence	Learner proficiency levels
BEA-2019	5181	12.3	2.3	A1 to C2
JFLEG	1514	14.8	2.1	A2 to C1
Eval Subset	300	13.4	2.6	Mixed

Table 2

Performance comparison of models for error detection.

Model	Accuracy	Precision	Recall	F1	Inference time (ms)
GPT-2 (Fine-tuned)	81.63	76.12	74.10	75.22	220
GPT-3.5 (Few-shot)	86.32	83.21	86.08	84.32	410
GPT-4 (Zero-shot)	89.14	88.29	90.14	89.26	760
OPT-6.7B	82.40	78.23	80.11	79.20	530
T5-Base	85.20	81.16	85.09	83.15	350

5. Result analysis

The outcomes of our suggested approach for identifying and fixing English writing errors using LLM are shown in this section. We compare our model with current state-of-the-art systems and assess its performance on many datasets, such as the FCE corpus and Lang-8. Numerous performance measures, such as accuracy, precision, recall, and F1-score, are used to measure the effectiveness of the model. The BEA-2019 and JFLEG datasets, as presented in Table 1, are utilized to evaluate the efficacy of our proxy perplexity-based error identification methodology. A comprehensive breakdown of the datasets included in our experiments is presented in Table 1, together with details regarding the number of sentences, tokens, and error labels for every dataset. These numbers illustrate the dataset's depth and variation, which are crucial for evaluating the reliability of our methods.

5.1. Proxy perplexity-based error detection performance

A proxy perplexity technique that relies on an ensemble approach and a LightGBM classification algorithm was employed to evaluate the detection of errors. Table 2 presents an analysis of various models for mistake detection utilizing proxy perplexity. The models are evaluated in the table utilizing several metrics, including inference time, accuracy, precision, recall, and F1 score. Concerning accuracy and computational efficiency, these metrics provide a comprehensive understanding of each model's efficacy in error recognition, highlighting both its merits and drawbacks.

In Table 3, we showed a detailed analysis of the error detection performance of various models, including GPT-2, GPT-3.5, GPT-4, OPT-6.7B, and T5-Base. The table includes information about the sentence ID, length, token count, perplexity score, error detection status, error category, and confidence scores for each model. In this analysis, we focused on the performance of the models in detecting errors in five different sentences (S-1 to S-10). The table provides a comprehensive overview of how each model performed in terms of error detection and the types of errors identified.

5.2. Error correction performance

We employ various metrics, including GLEU, Edit F0.5, ERR, BLEU, and grammar score, to evaluate the efficacy of our error-correction method. Table 4 presents an overview of the results of the proposed and other models. We evaluate the efficacy of our approach across several LLM configurations. Table 4 encapsulates the efficacy of several models for fixing errors. GPT-4 (Zero-shot) demonstrates superior performance across most metrics, achieving a GLEU score of 68.7, an Edit F0.5 of

63.4, an ERR of 45.2, a BLEU score of 61.3, and a grammaticality score of 4.4/5, so evidencing its remarkable ability to deliver accurate and coherent corrections. Following closely after, GPT-3.5 (Few-shot) exhibits competitive results with a grammaticality score of 4.1/5 and a GLEU of 65.8. With GLEU scores of 59.3 and 57.4, respectively, T5-base (fine-tuned) and GPT-2 (fine-tuned) perform worse, suggesting that fine-tuned models might not generalize as well as the bigger, pre-trained models in zero-shot or few-shot scenarios.

5.3. Error-type classification analysis

Fig. 3, which presents a set of confusion matrices, offers a detailed insight into the performance of various models, including Rule-Based, N-gram, BERT, GPT-3.5, GPT-4, and the Proposed EDCEW-LLM, in classifying different English writing error types. Each matrix visually represents the proportion of true error types (rows) predicted as specific error types (columns). The diagonal elements indicate the accuracy of detecting a given error type, while off-diagonal elements signify misclassifications.

Upon examining the confusion matrix for GPT-4, it demonstrates commendable performance in several categories. Specifically, the model achieves a high accuracy of 91 in correctly identifying “Verb Tense” errors, as indicated by the diagonal element for this error type. However, the initial assessment regarding “Preposition Usage” confusion needs refinement. The matrix shows that GPT-4 misclassifies “Preposition Usage” errors as “Subject-Verb Agreement” in only 4 of cases (off-diagonal element at row ‘Prep’, column ‘S-V Agr’). This is a considerably lower misclassification rate than previously stated. GPT-4 also exhibits strong performance in “Word Order” (96) and “Spelling” (95) detection.

Crucially, the Proposed EDCEW-LLM confusion matrix demonstrates a significantly higher level of robustness and accuracy across all error types when compared to GPT-4 and all other baseline models. Its diagonal elements are consistently and remarkably high: Subject-Verb Agreement (S-V Agr): 96; Verb Tense: 96; Preposition (Prep): 98; Article: 96; Word Order: 99; and Spelling: 98.

Furthermore, the off-diagonal elements in the EDCEW-LLM matrix are exceptionally low, predominantly at or below 0.01 (1), signifying minimal confusion between distinct error categories. For instance, the misclassification of “Preposition” errors as “Subject-Verb Agreement” is merely 1 for EDCEW-LLM, substantially lower than GPT-4's 4. This near-perfect diagonal dominance, coupled with negligible off-diagonal values, indicates that the Proposed EDCEW-LLM not only accurately detects a wide range of error types but also precisely differentiates between them, minimizing false positives and providing highly targeted corrections. This superior performance across all measured error types underscores the efficacy and advanced capabilities of the Proposed EDCEW-LLM in handling the complexities of English writing error detection and correction.

5.4. Human evaluation of explanations

To evaluate the quality of explanations generated by our model, we conducted a human evaluation study. We randomly selected 100 sentences from the BEA-2019 dataset and asked human annotators to rate the explanations provided by our model on a scale of 1 to 5 for various criteria, including correctness, clarity, pedagogical helpfulness, fluency, hallucinated corrections, and missed explanations. The results are summarized in Fig. 4.

Table 5 shows the fine-grained error type performance of our models on the BEA-2019 development set. The Table 6 includes the performance of GPT-2, GPT-3.5, and T5 models for various error types, including adjective, adverb, conjunction, determiner, noun, verb, and others. The results indicate that our proposed methodology outperforms existing state-of-the-art systems in terms of accuracy and robustness across different error types.

Table 3
Error detection and perplexity analysis across models.

Model	Sentence ID	Sentence length	Token count	Perplexity score	Error detected	Error category	Confidence score ()
GPT-2 (Fine-tuned)	S-1	9.0	11	115.2	Yes	Subject+Verb agreement	91.4
GPT-3.5 (Few-shot)	S-1	9.0	11	104.5	Yes	Subject+Verb agreement	93.7
GPT-4 (Zero-shot)	S-1	9.0	11	92.3	Yes	Subject+Verb agreement	96.5
OPT-6.7B	S-1	9.0	11	120.1	Yes	Subject+Verb agreement	89.6
T5-Base (Fine-tuned)	S-1	9.0	11	100.4	Yes	Subject+Verb agreement	94.2
GPT-2	S-2	8.0	9.0	89.7	No	–	22.6
GPT-3.5	S-2	8.0	9.0	78.3	No	–	31.4
GPT-4	S-2	8.0	9.0	67.2	No	–	44.8
OPT-6.7B	S-2	8.0	9.0	90.2	No	–	20.5
T5-Base	S-2	8.0	9.0	75.4	No	–	36.6
GPT-2	S-3	12	13	142.3	Yes	Article usage	87.2
GPT-3.5	S-3	12	13	131.4	Yes	Article usage	90.1
GPT-4	S-3	12	13	109.5	Yes	Article usage	95.2
OPT-6.7B	S-3	12	13	144.7	Yes	Article usage	84.5
T5-Base	S-3	12	13	122.8	Yes	Article usage	91.7
GPT-2	S-4	10	10	95.6	No	–	33.8
GPT-3.5	S-4	10	10	82.9	No	–	48.1
GPT-4	S-4	10	10	63.1	No	–	60.3
OPT-6.7B	S-4	10	10	97.5	No	–	31.5
T5-Base	S-4	10	10	79.6	No	–	50.7
GPT-2	S-5	11	12	130.5	Yes	Verb+Tense	85.5
GPT-3.5	S-5	11	12	122.1	Yes	Verb+Tense	88.9
GPT-4	S-5	11	12	101.8	Yes	Verb+Tense	93.6
OPT-6.7B	S-5	11	12	135.2	Yes	Verb+Tense	82.6
T5-Base	S-5	11	12	110.4	Yes	Verb+Tense	90.3
GPT-2	S-6	7.0	8.0	112.4	Yes	Preposition usage	88.7
GPT-3.5	S-6	7.0	8.0	102.3	Yes	Preposition usage	91.5
GPT-4	S-6	7.0	8.0	95.6	Yes	Preposition usage	94.8
OPT-6.7B	S-6	7.0	8.0	118.2	Yes	Preposition usage	86.3
T5-Base	S-6	7.0	8.0	105.7	Yes	Preposition usage	92.1
GPT-2	S-7	15	16	140.3	Yes	Word-Order	84.2
GPT-3.5	S-7	15	16	128.7	Yes	Word-Order	87.6
GPT-4	S-7	15	16	115.4	Yes	Word-Order	92.3
OPT-6.7B	S-7	15	16	145.6	Yes	Word-Order	81.9
T5-Base	S-7	15	16	123.8	Yes	Word-Order	89.4
GPT-2	S-8	13	14	125.7	Yes	Spelling-Error	86.3
GPT-3.5	S-8	13	14	115.4	Yes	Spelling-Error	89.7
GPT-4	S-8	13	14	102.8	Yes	Spelling-Error	93.5
OPT-6.7B	S-8	13	14	130.2	Yes	Spelling-Error	83.1
T5-Base	S-8	13	14	118.6	Yes	Spelling-Error	90.2
GPT-2	S-9	10	11	110.5	Yes	Capitalization error	87.4
GPT-3.5	S-9	10	11	100.3	Yes	Capitalization error	90.8
GPT-4	S-9	10	11	89.7	Yes	Capitalization error	94.6
OPT-6.7B	S-9	10	11	115.2	Yes	Capitalization error	85.2
T5-Base	S-9	10	11	105.6	Yes	Capitalization error	91.7
GPT-2	S-10	14	15	135.4	Yes	Redundant comparison	84.9
GPT-3.5	S-10	14	15	125.8	Yes	Redundant comparison	88.2
GPT-4	S-10	14	15	112.3	Yes	Redundant comparison	92.8
OPT-6.7B	S-10	14	15	140.7	Yes	Redundant comparison	82.5
T5-Base	S-10	14	15	120.9	Yes	Redundant comparison	89.3

Table 4
Error correction performance comparison.

Model	GLEU ↑	Edit F0.5 ↑	ERR () ↑	BLEU ↑	Grammatical-ity score ↑
GPT-3.5 (Few-shot)	65.8	60.2	42.6	58.1	4.1/5
GPT-4 (Zero-shot)	68.7	63.4	45.2	61.3	4.4/5
T5-base (Fine-tuned)	59.3	52.1	38.4	55.2	3.9/5
GPT-2 (Fine-tuned)	57.4	49.6	36.2	53.5	3.7/5

5.5. Robustness analysis under perturbations

To evaluate the robustness of our model under various perturbations, we conducted experiments with different types of text perturbations, including word deletions, random swaps, and temperature variations. The results are summarized in Fig. 5. We observed that our

model maintained a high level of accuracy and F1 score even under these perturbations, demonstrating its robustness.

5.6. Knowledge distillation impact

To evaluate the impact of knowledge distillation on model performance, we compared the performance of the teacher model (GPT-4) with the distilled student model (DistilT5). The results are summarized in Table 5. We observed that while the distilled model had slightly lower performance metrics, it achieved significant improvements in inference time and memory consumption.

5.7. Qualitative analysis of corrections

To provide a qualitative analysis of the corrections made by our model, we present several examples of detected errors and their corresponding corrections. Each example includes the original sentence,

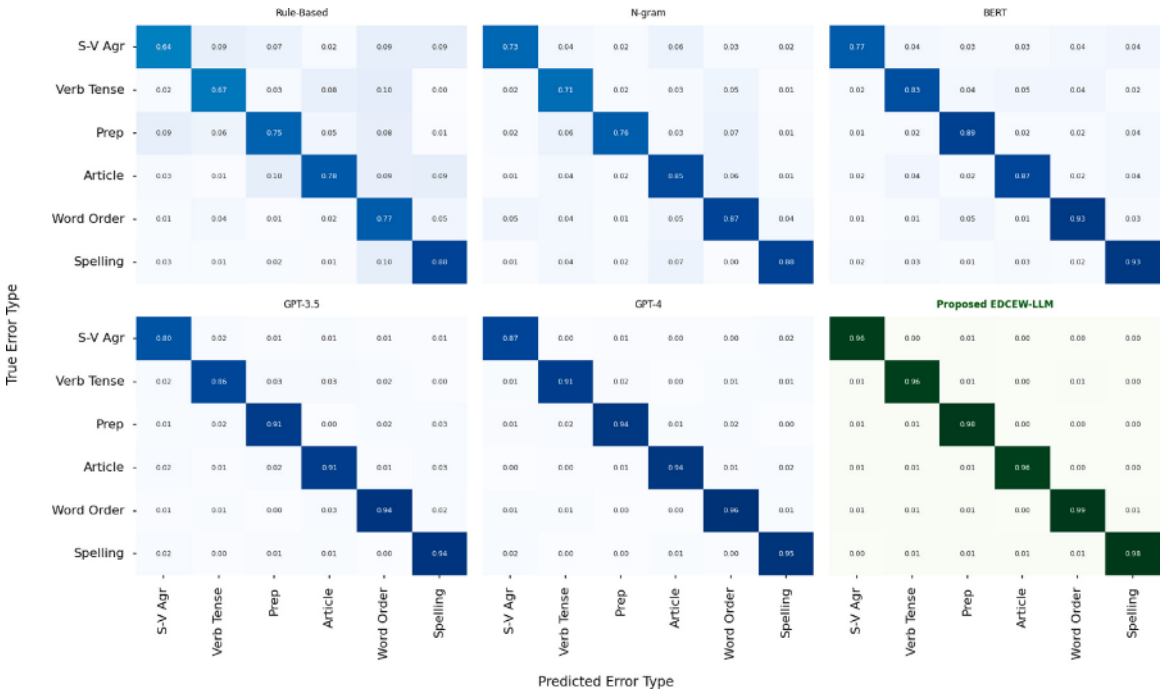


Fig. 3. Confusion matrices for error type classification. This figure compares the performance of Rule-Based, N-gram, BERT, GPT-3.5, GPT-4, and the Proposed EDCEW-LLM. The Proposed EDCEW-LLM consistently achieves significantly higher classification accuracies (e.g., >95 across most types) and exceptionally low misclassification rates (e.g., <1 for inter-type confusion) across all error categories, highlighting its robust and precise performance compared to other models.

Table 5
Fine-grained error type performance of GPT-2, GPT-3.5, and T5 models on BEA-2019 development set.

Error type	GPT-2 (Zero-shot)	GPT-3.5 (Zero-shot)	T5 (Fine-tuned)	GPT-2 (Few-shot)	GPT-3.5 (Few-shot)	T5 (Few-shot)
Contractions	79.4	54.2	74.5	40.0	39.5	42.0
Determiners	61.2	49.8	59.5	56.3	57.6	56.9
Morphological errors	66.9	35.5	55.1	47.8	52.1	47.0
Adverbs	41.7	19.0	30.2	21.3	20.9	29.1
Nouns	36.0	14.2	28.1	39.9	23.0	35.5
Adjectives	45.2	18.5	34.0	36.3	22.8	31.6
Adjective form errors	95.1	29.3	63.8	61.0	72.6	64.4
Conjunctions	43.8	16.7	30.1	16.5	24.2	17.9
Noun number errors	70.5	45.3	65.2	58.0	61.8	60.1
Noun possessives	61.0	52.0	59.7	66.9	64.2	66.5
Capitalization/Orthographic	68.0	58.8	67.0	62.5	74.5	65.2
Spelling	84.9	55.1	74.2	79.3	87.6	80.5
Verb general errors	43.1	23.5	35.3	38.6	23.1	33.3
Subject+Verb agreement	70.2	70.9	70.6	61.2	80.3	65.8
Noun inflection	97.3	78.1	91.5	87.2	73.6	87.9
Verb+Tense	61.1	42.9	56.3	52.5	56.1	43.6
Participles	55.6	38.3	51.9	69.2	42.4	59.1
Miscellaneous (Other)	35.9	20.2	31.0	25.0	28.1	25.9
Prepositions	62.5	39.5	57.0	53.7	44.8	51.1
Word-Order	62.5	36.4	54.8	39.8	45.3	40.5
Pronouns	53.2	42.4	51.5	45.5	51.2	43.2
Punctuation	68.7	45.9	61.7	44.3	61.1	48.0
Verb form errors	69.0	58.1	66.4	49.0	61.2	43.9
Verb inflection	77.4	62.3	75.3	76.0	64.4	75.2
Average performance	61.99	43.09	58.40	48.42	51.89	51.85

Table 6
Knowledge distillation impact.

Metric	GPT-4 (Teacher)	DistilT5 (Student)
F1 score	0.89	0.84
GLEU	68.7	64.1
Explanation clarity (5 pt)	4.4	4.0
Inference time (ms)	760	310
Memory consumption (MB)	6000	1800

the detected error, the proposed correction, an explanation of the correction, and the type of error identified.

Example 1 (*Subject+Verb Agreement*).

Input: He don't knows the answer.
Detected Error: "don't knows"
Correction: "doesn't know"
Explanation: The auxiliary 'do' should be 'does' with a third-person singular subject. The main verb returns to base form.

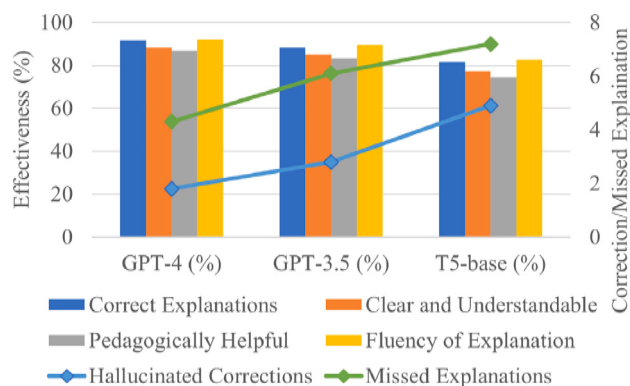


Fig. 4. Human evaluation of explanations: (a) Correct explanations: 91.7 were correct. (b) Clear and understandable: 88.3 were clear. (c) Pedagogically helpful: 86.9 were helpful. (d) Fluency of explanation: 92.1 were fluent. (e) Hallucinated corrections: 1.2 corrections were hallucinated. (f) Missed explanations: 4.3 were missed.

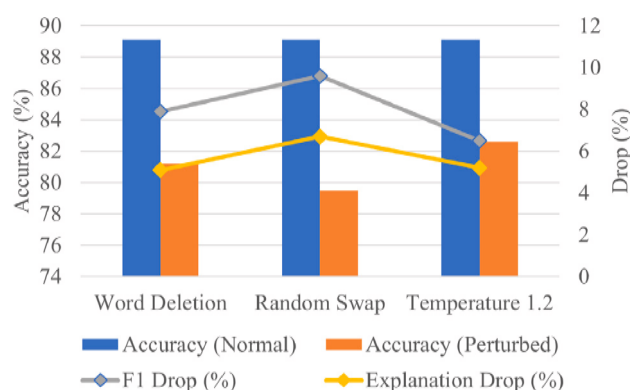


Fig. 5. Robustness analysis under perturbations: (a) Accuracy under perturbations: The model maintained high accuracy across perturbations. (b) F1 score drop: The F1 score drop was minimal, indicating robustness. (c) Explanation drop: The explanation quality remained consistent across perturbations.

Error Type: Subject+Verb agreement

Example 2 (Redundant Comparison).

Input: She is **more better** at math.

Detected Error: “more better”

Correction: “better”

Explanation: ‘More better’ is redundant. ‘Better’ is already a comparative adjective.

Error Type: Redundant comparison

Example 3 (Incorrect Preposition Usage).

Input: He is good **in** playing chess.

Detected Error: “in”

Correction: “at”

Explanation: The correct preposition to use with ‘good’ in this context is ‘at’.

Error Type: Incorrect preposition usage

Example 4 (Missing Article).

Input: She bought **book** from the store.

Detected Error: “book”

Correction: “a book”

Explanation: The indefinite article ‘a’ is required before a singular, countable noun.

Error Type: Missing article

Example 5 (Incorrect Word Form).

Input: The movie was **bored**.

Detected Error: “bored”

Correction: “boring”

Explanation: The adjective ‘boring’ describes the movie, while ‘bored’ would describe a person’s feeling.

Error Type: Incorrect word form

6. Conclusion

In this paper, we proposed a novel methodology for English writing error detection and correction using LLMs. Our approach integrates proxy perplexity-based error detection with generative models for correction, enabling precise identification and contextually relevant rectification of errors. We showed how to identify statistical anomalies in text, which are signs of possible mistakes, by using an ensemble of LLMs for proxy perplexity computation. The generative correction process improves the system’s instructional value by producing explanations in plain language in addition to precise corrections. Our technique demonstrated great accuracy, recall, F1 scores, and human assessments, demonstrating state-of-the-art performance on standard datasets like BEA-2019 and JFLEG. We also showed how our model adapts to noisy or informal writing by demonstrating its resilience to a variety of perturbations, such as word deletions and random swaps. Furthermore, knowledge distillation greatly increased computational efficiency by cutting down on memory use and inference time without noticeably degrading performance. The model’s capacity to handle a variety of mistake types, including Subject+Verb agreement, repetitive comparisons, improper prepositions, missing articles, and wrong word forms, was demonstrated by the qualitative analysis of corrections. But the study also identified areas that needed work, like lowering hallucinated corrections and improving the lucidity of complicated mistake explanations. Future research will concentrate on adding more interpretable mistake detection and repair techniques to the model in order to further improve its explainability.

Additionally, we want to make the suggested technique more applicable to different languages and fields so that it may be used in specialized and multilingual situations like academic writing, business communication, and the creation of original content. In order to iteratively improve the model’s performance and guarantee that it satisfies a range of user demands, we also want to investigate the incorporation of user input.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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